A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.3: Compression

Concept 1.3.2: Lossy vs Lossless Compression

Total Questions: 10

Mark Schemes begin on Page 7

Question 1 | Source: 9618_w25_qp_12 (Q6.a)

6 **(a)** A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression ....

Justification ....

....

....

.... [1]

Question 2 | Source: 9618_s25_qp_11 (Q2.b.ii)

2 Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

(ii) Identify whether the lossy or lossless compression method is more appropriate for real-time bit streaming. Justify your answer.

Compression method ....

Justification ....

....

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....

.... [3]

Question 3 | Source: 9618_w24_qp_12 (Q7.a.iii)

7 A student takes a photograph of a science experiment.

(a) The photograph is saved as a bitmapped image.

(iii) Identify **one** lossless method of compressing an image.

.... [1]

Question 4 | Source: 9618_w24_qp_13 (Q6.b.iii)

6 A computer system stores text, images and sound.

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

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.... [2]

Question 5 | Source: 9618_s24_qp_13 (Q2.b.i)

2 A photograph is stored as a bitmap image.

(b) The photograph is compressed before being uploaded to a web server.

(i) Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

1 ....

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2 ....

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3 ....

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[3]

Question 6 | Source: 9618_w23_qp_12 (Q6.a)

6 **(a)** A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

Lossy	**Lossless**

Justification ....

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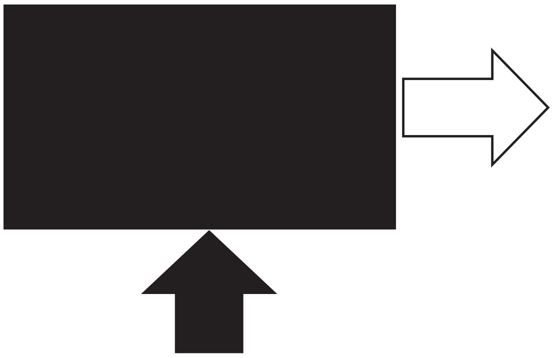
.... [3]

Question 7 | Source: 9618_w22_qp_12 (Q8.c.ii)

8 The following bitmap image has a resolution of 4096 × 4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920 × 1080 pixels.

(c) A second bitmap image is stored using a colour depth of 8 bits per pixel.

The file is compressed using run-length encoding (RLE).

(ii) RLE is an example of lossless compression.

Explain why lossless compression is more appropriate than lossy compression for a text file.

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.... [2]

Question 8 | Source: 9618_w21_qp_11 (Q7.b.ii)

7 Bobby is recording a sound file for his school project.

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

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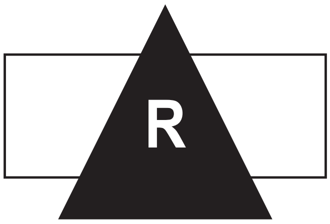
.... [2]

Question 9 | Source: 9618_w21_qp_12 (Q5.b.ii)

5 Riya has created the following logo as a vector graphic.



(b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

(ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1 ....

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Method 2 ....

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[4]

Question 10 | Source: 9618_s21_qp_11 (Q1.b)

1 Anya scans an image into her computer for a school project.

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method ....

Description ....

....

....

....

....

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q6.a)

6(a)	**1 mark** for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	**1**

Mark Scheme for Question 2 | Source: 9618_s25_ms_11 (Q2.b.ii)

2(b)(ii)	No mark for choice but lossy is **most** appropriate **1 mark** each to **max 3** e.g. • Reduces file size more than lossless • so significantly less bandwidth / data is needed • so buffering is reduced even more than with lossless • Data can be removed which cannot be seen • reducing quality without impacting experience • for example, because resolution of video can be reduced // sample rate of audio can be reduced	**3**

Mark Scheme for Question 3 | Source: 9618_w24_ms_12 (Q7.a.iii)

7(a)(iii)	**1 mark** for correct method For example: • Run-Length Encoding	**1**

Mark Scheme for Question 4 | Source: 9618_w24_ms_13 (Q6.b.iii)

6(b)(iii)	**1 mark** for each bullet point (**max 2**) e.g. • The change may not be noticeable // Data removed is usually not noticed by the human eye • … for example, changes in shade/detail • It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	**2**

Mark Scheme for Question 5 | Source: 9618_s24_ms_13 (Q2.b.i)

2(b)(i)	**1 mark** each to **max 3**: • The file takes less storage space on the web server than if lossless compression was used • The file is faster to upload/download to/from the server than if lossless compression was used • The file uses less bandwidth to transmit than if lossless compression was used • The file consumes less data allowance than if lossless compression was used	**3**

Mark Scheme for Question 6 | Source: 9618_w23_ms_12 (Q6.a)

6(a)	**1 mark** for each bullet point (**max 3**) Lossy compression (ticked) • Loss of quality will not be noticed • Needs to be viewed in real time so less bandwidth needed if file size smaller • Smaller file sizes will reduce buffering so the video will play more smoothly • Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) • Original recording may not have been made in high resolution • Could be streaming to high bandwidth devices • The reduction in the file size is sufficient for the receiving device • Viewers do not want any loss of quality	**3**

Mark Scheme for Question 7 | Source: 9618_w22_ms_12 (Q8.c.ii)

8(c)(ii)	**1 mark** for each bullet point: • all the data is required // no data can be lost • ... otherwise text file will be corrupted / not make sense	**2**

Mark Scheme for Question 8 | Source: 9618_w21_ms_11 (Q7.b.ii)

7(b)(ii)	**1 mark** per bullet point to **max 2** e.g. • Reduce amplitude to only the range used • … limited amplitudes mean fewer bits per sample • Run-length-encoding • … Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	**2**

Mark Scheme for Question 9 | Source: 9618_w21_ms_12 (Q5.b.ii)

5(b)(ii)	**1 mark** for each bullet point to **max 2** for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	**4**

Mark Scheme for Question 10 | Source: 9618_s21_ms_11 (Q1.b)

1(b)	**1 mark** for naming method, **1 mark** per description to **max 2** • Run-length encoding • Replace **sequences** of the **same colour** pixel • … with colour code and number of identical pixels	**3**